GazePoint

Team Name: **Team GazeConnect**

Project Title: GazePoint - An Intuitive Eye-Controlled Mouse

Domain: Assistive Technology & Human-Computer Interaction (HCI)

Theme: Accessibility & Empowerment

Category: Software/Hardware

Problem Statement

- **The Input Gap:** Individuals with severe motor disabilities (like ALS, SCI, or Cerebral Palsy) cannot use standard mice or keyboards.
- **Barriers to Independence:** This gap creates immense barriers to communication, education, employment, and personal autonomy.
- **Isolation:** Lack of digital access leads to social isolation and a profound loss of connection with the world.
- **Fatigue:** Existing alternative inputs (like sip-and-puff or head switches) can be slow, unintuitive, and physically exhausting.

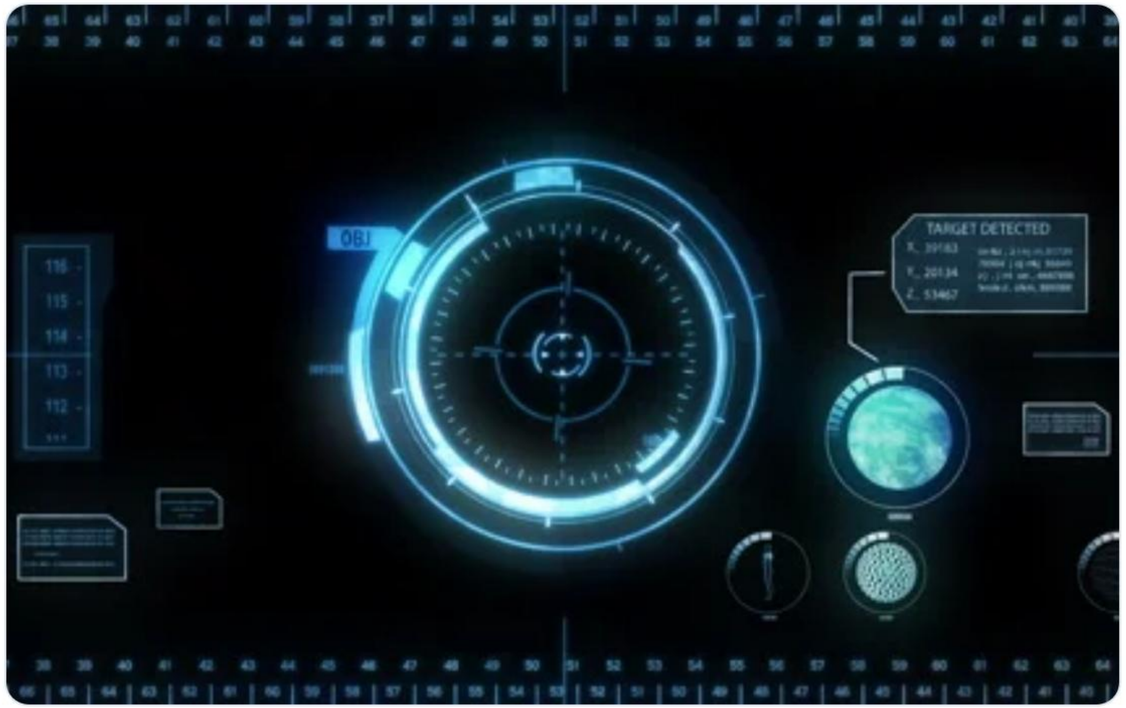
Proposed Solution: GazePoint

How It Works

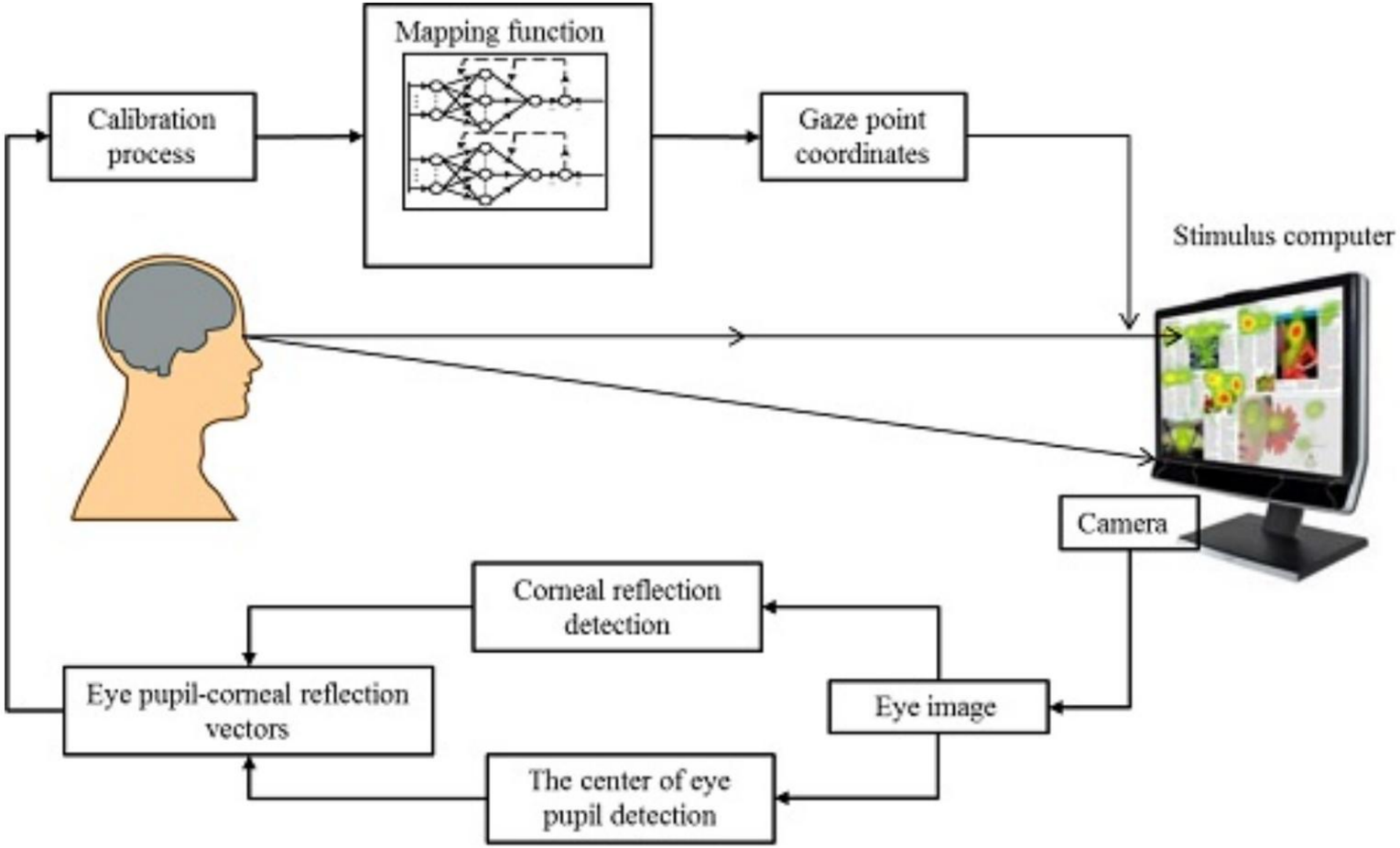
- An AI-powered camera tracks the user's eye movements.
- The system translates the user's gaze into a real-time cursor position on the screen.
- A "click" is performed by dwelling (staring) on a button for a set time (e.g., 0.5 seconds) or by blinking.

Key Features

- Complete hands-free computer control
- Intuitive and low-fatigue interaction
- Full digital access (browsing, typing, etc.)



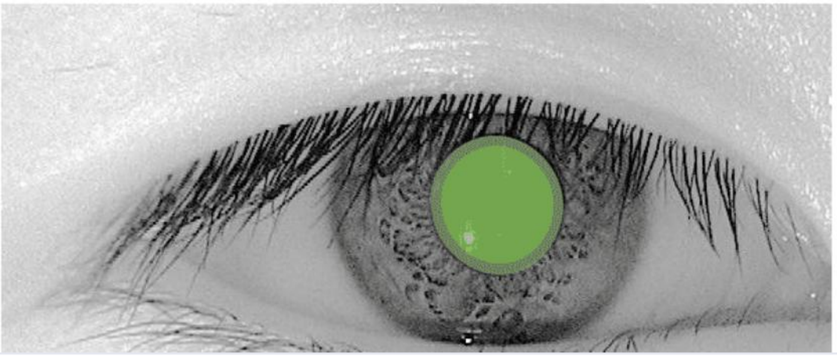
Technical Approach: System Flow



Technical Approach: Core Technologies

- **Image Acquisition:** A standard high-resolution webcam (or IR camera for higher accuracy) captures the user's face.
- **Face & Eye Detection:** Use OpenCV & Dlib libraries to find facial landmarks and isolate the eye regions.
- **Gaze Estimation:** An algorithm calculates the pupil's center relative to the eye corners to determine gaze direction.
- **Cursor Mapping:** Translate the gaze vector into on-screen (X, Y) pixel coordinates.
- **Click Emulation:** A timer module detects "dwell time" or a "blink" to trigger a mouse click event.

Iris illuminated by infrared light.



Feasibility

Technical Feasibility

The core technologies (OpenCV, Dlib, Python) are mature, open-source, and well-documented. Proofs-of-concept are widely available.

Operational Feasibility

The system is non-invasive and can run on most modern computers. A software-only solution (using a standard webcam) makes it highly accessible.

```
import numpy as np
import cv2 as cv
from matplotlib import pyplot as plt
img = cv.imread('road.jpg',0)
edges = cv.Canny(img,100,200)
plt.subplot(121),plt.imshow(img,cmap = 'gray')
plt.title('Original Image'), plt.xticks([]),
plt.yticks([]),plt.subplot(122),plt.imshow(edges,cmap = 'gray')
plt.title('Edge Image'), plt.xticks([]), plt.yticks([])
plt.show()
```

Challenges & Solutions

Challenges

- **"Midas Touch" Problem:** Accidentally clicking everything you look at.
- **Accuracy & Drift:** Small head movements causing the cursor to lose calibration.
- **Cost:** High-end IR systems are prohibitively expensive.

Solutions

- Implement robust **Dwell Time** (stare to click) and dedicated "blink" commands.
- Use software filters to smooth jitter and add a quick, hotkey-based **re-calibration** function.
- Focus on a **software-only solution** using standard webcams to ensure accessibility.

Impact & Benefits

- **Restores Communication:** Enables users to type, send emails, and use speech-generating (AAC) devices.
- **Enables Education & Employment:** Unlocks access to online learning, coding, writing, and professional work.
- **Enhances Independence:** Allows users to browse the web, control smart home devices, and interact with the world on their own terms.
- **Reduces Fatigue:** Provides a less physically taxing and more intuitive input method.

Research and References

- **Core Technology:** Built on foundational computer vision libraries like **OpenCV** and **Dlib** for facial landmark detection.
- **Hardware & Gaze Models:** Research inspired by industry leaders in gaze interaction, such as **Tobii Dynavox**.
- **Problem Validation:** Studies and reports from organizations like the **ALS Association** and **WHO** on "locked-in" syndrome and the need for assistive tech.

Questions?

Image Sources

https://static.vecteezy.com/system/resources/previews/070/130/936/non_2x/eye-tracking-clean-frame-object-free-vector.jpg

Source: www.vecteezy.com

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https://i2-prod.gazettelive.co.uk/incoming/article30270360.ece/ALTERNATES/s615/0_Disabled-person-frustrated-with-their-finances.jpg

Source: www.gazettelive.co.uk

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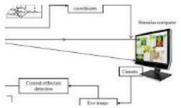
https://images.pond5.com/digital-hud-blue-futuristic-computer-footage-054839280_iconl.jpeg

Source: www.pond5.com



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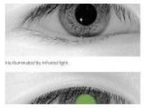
Source: www.mdpi.com



https://imotions.com/wp-content/uploads/2022/10/infrared-eye-trackers-1.png

Source: imotions.com

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Source: www.projectpro.io

Image Sources

https://img.freepik.com/premium-photo/african-smiling-man-sitting-wheelchair-using-laptop-computer-mall_249974-6192.jpg

Source: www.freepik.com

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Source: www.swcil.com

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Source: www.freepik.com